

Fixed-Income Team Projects

BUS 411

Table of contents

Common Scope and Standards	2
1 Lecture 1: Bond Features, Structures, Markets, and Risks	3
1.1 Project 1: Read the Bond Before Recommending It	3
1.2 Project 2: Funding Choice for a Corporate Treasurer	3
2 Lecture 2: Bond Pricing, Yield Measures, and Total Return	4
2.1 Project 3: Price, Yield, and Horizon Return Case	4
2.2 Project 4: Premium, Par, or Discount?	4
3 Lecture 3: Duration, Convexity, and Portfolio Strategies	5
3.1 Project 5: Immunize a Known Liability	5
3.2 Project 6: When Convexity Earns Its Keep	6
4 Lecture 4: Benchmark Spreads and Term Structure	6
4.1 Project 7: Build and Read a Government Yield Curve	6
4.2 Project 8: Bootstrap a Discount Curve by Hand	7
5 Lecture 5: Treasury and Corporate Debt Markets	7
5.1 Project 9: On-the-Run Liquidity and Treasury Specialness	7
5.2 Project 10: Credit Event Anatomy	8
6 Lecture 6: Bond Portfolio Construction	8
6.1 Project 11: Build a Practical Bond Index Sample	8
6.2 Project 12: Rebalance Under Real-World Constraints	9
7 Lecture 7: Interest-Rate Models	9
7.1 Project 13: Vasicek as a Risk Scenario Generator	9
7.2 Project 14: Why Hull–White Must Fit Today’s Curve	10

8	Lecture 8: Embedded Options and Option-Adjusted Measures	10
8.1	Project 15: Callable Bond Relative-Value Review	10
8.2	Project 16: Backward Induction in an Interest-Rate Tree	11
9	Lecture 9: Interest-Rate Futures	11
9.1	Project 17: Hedge a Bond Position with Treasury Futures	11
9.2	Project 18: Read the SOFR Futures Curve	12
10	Lecture 10: Credit Spread Exposure and Credit Risk Modeling	12
10.1	Project 19: Spread Duration and Credit Scenario Budget	12
10.2	Project 20: Expected Loss and Bondholder Recovery	13
11	Lecture 11: Interest-Rate Options	13
11.1	Project 21: Cap, Floor, or Swap for a Borrower?	13
11.2	Project 22: Option on Treasury Futures Event Hedge	14

Common Scope and Standards

Each project is designed for a team of **two or three students**. The final product is a **10–12 minute team presentation** supported by a concise analyst-style exhibit package. All team members should participate in the analysis and presentation.

- **Financial reasoning:** Emphasize interpretation, assumptions, tradeoffs, and recommendations. Coding is optional and should remain limited to simple data retrieval, calculations, or charts. A spreadsheet is sufficient for every project.
- **Reproducibility:** State the observation date, units, conventions, formulas, and transformations. Submit the source data or a clean extract, plus the spreadsheet or short script used to produce the exhibits.
- **Attribution:** Cite every external dataset, market quote, filing, figure, and factual claim. Distinguish observed data from assumptions and estimates.
- **Scope discipline:** Use a manageable sample and explain exclusions. When live or historical access is difficult, teams may use an instructor-provided dated snapshot of the named source.
- **Professional judgment:** Identify limitations and discuss how they affect the conclusion. These projects are educational analyses, **not investment advice**.
- **Presentation standard:** Use legible figures, label axes and units, define fixed-income terminology, and end with a clear conclusion. Appendices may contain supporting calculations but are not additional presentation time.

Unless a project says otherwise, submit: (1) presentation slides in PDF, (2) a reproducible exhibit package, and (3) a one-page source and assumptions note.

1 Lecture 1: Bond Features, Structures, Markets, and Risks

1.1 Project 1: Read the Bond Before Recommending It

Scenario. A wealth-management team is choosing between two recently issued bonds from the same or comparable issuers. Your team must explain how contractual features change the investor's cash flows and risks.

Objectives

- Extract and compare coupon, maturity, seniority, security, call, sinking-fund, and covenant terms.
- Map each feature to interest-rate, credit, liquidity, reinvestment, inflation, and optional-ity risk.
- Recommend which bond better fits a stated conservative-income mandate.

Data sources. Issuer prospectus supplements and annual reports filed on **SEC EDGAR** or **SEDAR+**; **FINRA Fixed Income Data/Bond Center** for U.S. corporate bond reference information and recent trades; an instructor-provided prospectus snapshot if needed.

Deliverables. A one-page term sheet comparison; a risk map with at least six feature-to-risk links; a mandate-based recommendation with two limitations; and a 10–12 minute team presentation.

Rubric (100 points). Contract-term accuracy, 25; feature-to-cash-flow analysis, 25; risk reasoning, 25; recommendation and limitations, 15; presentation clarity and source attribution, 10.

1.2 Project 2: Funding Choice for a Corporate Treasurer

Scenario. A corporate treasurer must raise medium-term funding and is considering a fixed-rate bond, floating-rate note, or bank loan.

Objectives

- Compare cash-flow uncertainty, refinancing risk, covenants, seniority, and investor base across the three structures.
- Relate the choice to the firm's asset duration and rate outlook without relying on a directional forecast alone.
- Propose a defensible funding structure and identify the conditions under which it should change.

Data sources. A public issuer's latest **annual report**, debt footnote, and investor presentation; benchmark rates from **FRED** or the **Bank of Canada**; sample loan or note terms from issuer filings; instructor-provided terms where loan details are private.

Deliverables. A funding-needs diagnosis; a three-column structure comparison; one simple annual interest-cost scenario table; a recommendation and trigger conditions; and a 10–12 minute team presentation.

Rubric (100 points). Issuer and funding-needs analysis, 20; structure comparison, 30; scenario design and calculations, 20; recommendation and contingencies, 20; presentation and documentation, 10.

2 Lecture 2: Bond Pricing, Yield Measures, and Total Return

2.1 Project 3: Price, Yield, and Horizon Return Case

Scenario. An investor may buy a coupon bond but expects to sell it before maturity. Evaluate whether yield to maturity is an adequate decision measure.

Objectives

- Reproduce the quoted bond price from promised cash flows and yield conventions.
- Calculate current yield, yield to maturity, and a two- or three-year horizon return.
- Show how reinvestment rates and the sale yield affect realized return.

Data sources. **U.S. Treasury** or **Bank of Canada** bond data for an option-free government bond; **FINRA Fixed Income Data/TRACE** for a corporate alternative; official security terms from TreasuryDirect, the Bank of Canada, or an issuer filing.

Deliverables. A cash-flow timeline; transparent price and yield calculations; a three-scenario horizon-return table; an explanation of why realized return differs from YTM; and a 10–12 minute team presentation.

Rubric (100 points). Cash-flow and convention accuracy, 20; price and yield calculations, 25; horizon-return scenarios, 30; interpretation, 15; presentation and reproducibility, 10.

2.2 Project 4: Premium, Par, or Discount?

Scenario. A client believes that a discount bond is automatically a better bargain than a premium bond. Test the claim using matched securities.

Objectives

- Select two government or high-quality corporate bonds with similar maturity and credit exposure but different coupons.
- Explain the relationship among coupon rate, market yield, price, pull to par, and accrued interest.
- Compare after-tax or accounting considerations qualitatively where relevant.

Data sources. U.S. Treasury marketable-security data or Bank of Canada benchmark bond data; FINRA Fixed Income Data for corporate examples; official tax-agency guidance or accounting-policy disclosures for qualitative context.

Deliverables. A matched-bond table; clean- and dirty-price reconciliation for one settlement date; a price path or pull-to-par exhibit under a constant-yield assumption; a client-facing conclusion; and a 10–12 minute team presentation.

Rubric (100 points). Security matching and data quality, 20; pricing and accrued-interest work, 30; pull-to-par analysis, 20; financial explanation and caveats, 20; presentation quality, 10.

3 Lecture 3: Duration, Convexity, and Portfolio Strategies

3.1 Project 5: Immunize a Known Liability

Scenario. A small institution must fund a known payment in five to eight years using two or three government bonds.

Objectives

- Match present value and duration between the asset portfolio and the liability.
- Explain why duration matching provides only local protection.
- Stress-test the position under parallel and simple nonparallel yield-curve changes.

Data sources. U.S. Treasury Daily Treasury Par Yield Curve Rates, TreasuryDirect, or Bank of Canada benchmark yields and bond terms; an instructor-provided set of bond prices and cash flows if necessary.

Deliverables. Liability assumptions; bond selection and weights; present-value, duration, and convexity calculations; at least three rate-shock results; a rebalancing note; and a 10–12 minute team presentation.

Rubric (100 points). Liability framing, 15; immunization calculations, 30; shock analysis, 25; interpretation and rebalancing judgment, 20; presentation and auditability, 10.

3.2 Project 6: When Convexity Earns Its Keep

Scenario. Two bonds have similar duration but different convexity. A portfolio manager asks when paying for greater convexity is worthwhile.

Objectives

- Construct or identify a duration-matched bond pair or small portfolios.
- Compare duration-only and duration-plus-convexity price estimates with full repricing.
- Explain the role of shock size, direction, yield volatility, and carry.

Data sources. Government bond terms and yields from the **U.S. Treasury** or **Bank of Canada**; prices from official sources or **FINRA Fixed Income Data**; instructor-provided option-free bond snapshots are acceptable.

Deliverables. A duration-matched comparison; repricing results for at least five yield shocks; an approximation-error chart; a discussion of convexity benefit versus likely cost; and a 10–12 minute team presentation.

Rubric (100 points). Matching design, 20; duration and convexity calculations, 25; full-repricing comparison, 25; economic interpretation, 20; presentation and source note, 10.

4 Lecture 4: Benchmark Spreads and Term Structure

4.1 Project 7: Build and Read a Government Yield Curve

Scenario. Prepare a morning curve briefing for a fixed-income desk on two dates with meaningfully different market conditions.

Objectives

- Construct comparable government yield curves and calculate key slopes.
- Separate changes in level, slope, and curvature.
- Link curve movement cautiously to contemporaneous policy and macroeconomic information.

Data sources. **FRED** Treasury constant-maturity series, **U.S. Treasury Daily Treasury Par Yield Curve Rates**, or **Bank of Canada** selected bond yields; policy releases from the **Federal Reserve** or **Bank of Canada**.

Deliverables. Two dated curve plots; 2s10s, 5s30s, and one curvature measure; a change decomposition; a short event timeline; a desk briefing conclusion; and a 10–12 minute team presentation.

Rubric (100 points). Data consistency and curve construction, 25; slope and curvature analysis, 25; event interpretation, 20; risk implications, 20; presentation and attribution, 10.

4.2 Project 8: Bootstrap a Discount Curve by Hand

Scenario. A valuation team needs discount factors and spot rates from a small set of coupon-bearing government securities.

Objectives

- Explain the sequential logic of bootstrapping.
- Derive discount factors and spot rates using four to six instruments.
- Validate the curve by repricing the input instruments and discuss interpolation risk.

Data sources. An instructor-provided internally consistent Treasury or Government of Canada price snapshot; alternatively, official bill and bond terms from the **U.S. Treasury** or **Bank of Canada** with same-date prices from a documented source.

Deliverables. A cash-flow matrix; step-by-step bootstrap calculations; discount-factor and spot-rate curves; repricing errors; an interpolation and convention note; and a 10–12 minute team presentation.

Rubric (100 points). Instrument and convention setup, 20; bootstrap accuracy, 35; repricing validation, 20; interpolation and limitation analysis, 15; presentation and reproducibility, 10.

5 Lecture 5: Treasury and Corporate Debt Markets

5.1 Project 9: On-the-Run Liquidity and Treasury Specialness

Scenario. A relative-value analyst observes that two similar Treasury securities trade at different yields and must explain why.

Objectives

- Compare an on-the-run security with a nearby off-the-run security.
- Distinguish duration and cash-flow differences from liquidity, benchmark demand, and repo specialness.
- Assess whether the yield difference is compensation for credit risk.

Data sources. **U.S. Treasury** auction results and security details; Treasury yield and price data from official releases or **FRED**; **Federal Reserve Bank of New York** repo data and publications; an instructor-provided same-date quote snapshot.

Deliverables. A matched-security profile; yield and duration comparison; a qualitative decomposition of the spread; a repo-market evidence exhibit; a relative-value conclusion with limitations; and a 10–12 minute team presentation.

Rubric (100 points). Security matching, 20; yield and risk comparison, 25; liquidity and repo reasoning, 30; conclusion and limitations, 15; presentation and sourcing, 10.

5.2 Project 10: Credit Event Anatomy

Scenario. A corporate issuer experiences a downgrade, earnings shock, acquisition, or refinancing event. Explain how bondholders were affected.

Objectives

- Build a dated event timeline using public information.
- Compare bond spread or price behavior before and after the event.
- Separate Treasury-rate effects from issuer-specific spread effects and discuss expected recovery or covenant implications.

Data sources. **FINRA TRACE** transaction data or FINRA bond market information; issuer filings and releases from **SEC EDGAR** or **SEDAR+**; rating-agency public releases; matched Treasury yields from **FRED**.

Deliverables. An event timeline; bond and benchmark exhibit; approximate rate-versus-spread price attribution; covenant, seniority, or recovery discussion; a bondholder conclusion; and a 10–12 minute team presentation.

Rubric (100 points). Event research and chronology, 20; market-data analysis, 25; rate/spread attribution, 25; credit reasoning, 20; presentation and citations, 10.

6 Lecture 6: Bond Portfolio Construction

6.1 Project 11: Build a Practical Bond Index Sample

Scenario. A small fund cannot hold every security in a broad bond index and needs a transparent sampling portfolio.

Objectives

- Design cells based on maturity, sector, quality, and duration.
- Select a feasible sample that resembles the benchmark's major exposures.
- Evaluate tracking-error sources, liquidity, turnover, and transaction-cost tradeoffs.

Data sources. Public fact sheets and holdings for a broad bond ETF or index fund; issuer and security data from fund disclosures and **FINRA Fixed Income Data**; instructor-provided benchmark weights where full index constituents are unavailable.

Deliverables. A benchmark exposure table; cell definitions; a 10- to 20-bond proposed sample or stylized sample; exposure gaps; a qualitative tracking-error budget; and a 10–12 minute team presentation.

Rubric (100 points). Benchmark diagnosis, 20; cell design, 25; portfolio selection and exposure matching, 25; implementation tradeoffs, 20; presentation and documentation, 10.

6.2 Project 12: Rebalance Under Real-World Constraints

Scenario. A bond portfolio has drifted outside duration, sector, issuer, and liquidity limits. The manager can make no more than five trades.

Objectives

- Diagnose active exposures relative to a benchmark or policy portfolio.
- Choose trades that reduce unintended risk while preserving an intentional view.
- Account for lot sizes, bid-ask costs, liquidity, and turnover.

Data sources. Public bond ETF holdings and fact sheets; **FINRA Fixed Income Data/TRACE** for trading activity; issuer terms from filings; an instructor-provided portfolio, benchmark, and bid-ask snapshot.

Deliverables. Before-trade exposure dashboard; proposed trades and rationale; after-trade exposures; estimated transaction cost; rejected alternatives; and a 10–12 minute team presentation.

Rubric (100 points). Exposure diagnosis, 25; trade design, 25; constraint and cost treatment, 25; before/after decision quality, 15; presentation and reproducibility, 10.

7 Lecture 7: Interest-Rate Models

7.1 Project 13: Vasicek as a Risk Scenario Generator

Scenario. A risk committee wants to know what a simple mean-reverting short-rate model can and cannot say about future rates.

Objectives

- Estimate or choose defensible Vasicek parameters from a historical short-rate series.
- Explain mean reversion, long-run mean, volatility, and the possibility of negative rates.
- Compare model-implied one-year rate distributions with historical changes.

Data sources. Effective federal funds rate or Treasury bill rates from **FRED**; overnight or Treasury bill rates from the **Bank of Canada**; an instructor-provided clean monthly series and parameter template.

Deliverables. Parameter rationale; a spreadsheet simulation or instructor-provided simulation output; a distribution and three sample paths; historical comparison; a model-use and model-risk memo; and a 10–12 minute team presentation.

Rubric (100 points). Parameter reasoning, 25; model interpretation, 25; scenario analysis, 20; validation and model-risk discussion, 20; presentation and reproducibility, 10.

7.2 Project 14: Why Hull–White Must Fit Today’s Curve

Scenario. A colleague proposes using a historically estimated equilibrium model to price a callable bond without first matching current market prices.

Objectives

- Explain the difference between equilibrium and no-arbitrage models.
- Demonstrate why fitting today’s discount curve matters for time-0 valuation.
- Show qualitatively how mean reversion and volatility affect future rate states and option value.

Data sources. Current discount factors or zero rates from **U.S. Treasury**, **FRED**, or **Bank of Canada** data; a small instructor-provided Hull–White tree or calibration output; public swaption or cap-volatility commentary if available.

Deliverables. A model-comparison table; a curve-fit or repricing exhibit; two parameter-sensitivity scenarios; a recommendation for plain versus option-embedded bonds; and a 10–12 minute team presentation.

Rubric (100 points). Model distinction, 25; curve-fit reasoning, 25; sensitivity interpretation, 20; instrument-specific recommendation, 20; presentation and attribution, 10.

8 Lecture 8: Embedded Options and Option-Adjusted Measures

8.1 Project 15: Callable Bond Relative-Value Review

Scenario. A portfolio manager must choose between a callable corporate bond and an option-free or weakly callable alternative from a comparable issuer.

Objectives

- Compare yield to worst, static spread, call schedule, and reinvestment risk.
- Decompose the callable bond conceptually into an option-free bond minus the issuer’s call option.
- Evaluate rate and volatility scenarios without treating the larger quoted spread as free compensation.

Data sources. **FINRA Fixed Income Data/TRACE**; issuer prospectus supplements from **SEC EDGAR** or **SEDAR+**; benchmark curves from **FRED**, the **U.S. Treasury**, or the **Bank of Canada**; instructor-provided OAS snapshots if needed.

Deliverables. A matched-bond term sheet; call timeline; yield/spread comparison; three rate-volatility scenarios; a mandate-based recommendation; and a 10–12 minute team presentation.

Rubric (100 points). Bond and call-term accuracy, 20; spread and yield analysis, 25; optionality reasoning, 30; recommendation and caveats, 15; presentation and sourcing, 10.

8.2 Project 16: Backward Induction in an Interest-Rate Tree

Scenario. A valuation committee wants an auditable explanation of how callability or putability changes a bond's value.

Objectives

- Price an option-free and callable or putable bond in a small recombining tree.
- Apply the exercise rule at each eligible node and calculate embedded-option value.
- Test sensitivity to volatility, call or put price, and spread.

Data sources. Instructor-provided short-rate tree and security terms; a same-date benchmark curve from the **U.S. Treasury** or **Bank of Canada** for context; issuer filings for a realistic exercise schedule.

Deliverables. A node-by-node spreadsheet; option-free and option-adjusted values; exercise-node map; at least three sensitivities; an audit note explaining timing conventions; and a 10–12 minute team presentation.

Rubric (100 points). Tree and cash-flow setup, 20; backward-induction accuracy, 30; exercise logic, 20; sensitivity and interpretation, 20; presentation and auditability, 10.

9 Lecture 9: Interest-Rate Futures

9.1 Project 17: Hedge a Bond Position with Treasury Futures

Scenario. A portfolio manager wants to reduce the rate exposure of a Treasury or high-quality corporate bond position for one month without selling the bonds.

Objectives

- Estimate the position's dollar duration or DV01.
- Select an appropriate Treasury futures contract and calculate a hedge ratio.
- Explain conversion factors, cheapest-to-deliver uncertainty, basis risk, and hedge slippage.

Data sources. **CME Group** Treasury futures contract specifications, conversion factors, and settlement data; **U.S. Treasury** yields and security terms; **FINRA** data for a corporate cash position; instructor-provided conversion-factor and cheapest-to-deliver snapshots.

Deliverables. Cash-position risk measure; contract choice; hedge-ratio calculation; two nonparallel-rate scenarios; implementation and basis-risk note; and a 10–12 minute team presentation.

Rubric (100 points). Cash-position risk measurement, 20; futures selection and hedge ratio, 30; delivery and basis-risk analysis, 25; scenario interpretation, 15; presentation and sources, 10.

9.2 Project 18: Read the SOFR Futures Curve

Scenario. A macro fixed-income desk needs a briefing on the policy path implied by the SOFR futures strip and how that view changed around a central-bank announcement.

Objectives

- Convert selected SOFR futures prices into implied forward rates.
- Compare strips on two dates and distinguish timing from magnitude of expected policy changes.
- Explain why futures-implied rates are not pure forecasts.

Data sources. CME Group Three-Month SOFR futures settlements and contract specifications; Federal Reserve policy statements and meeting calendar; effective federal funds rate from FRED; instructor-provided settlement snapshots if historical access is limited.

Deliverables. Two implied-rate strips; contract-month change table; policy-event timeline; interpretation of term or risk premiums and limitations; a desk conclusion; and a 10–12 minute team presentation.

Rubric (100 points). Contract and price conversion accuracy, 25; curve-change analysis, 25; policy interpretation, 20; futures-versus-forecast caveats, 20; presentation and reproducibility, 10.

10 Lecture 10: Credit Spread Exposure and Credit Risk Modeling

10.1 Project 19: Spread Duration and Credit Scenario Budget

Scenario. A credit portfolio has a yield advantage over government bonds, but the investment committee wants to know how much spread widening would erase one year of carry.

Objectives

- Estimate spread duration and annual carry for a bond or small portfolio.
- Calculate break-even spread widening and scenario losses.
- Distinguish spread risk from default loss, Treasury-rate risk, and liquidity risk.

Data sources. **FINRA Fixed Income Data/TRACE** and issuer filings; option-adjusted spread indexes from **FRED**; matched Treasury data from the **U.S. Treasury**; instructor-provided bond analytics where OAS is not freely available.

Deliverables. Position and benchmark profile; carry and spread-duration calculations; break-even widening; mild, severe, and recession scenarios; a risk-budget recommendation; and a 10–12 minute team presentation.

Rubric (100 points). Input and convention quality, 20; carry and break-even calculations, 30; scenario construction, 20; risk decomposition and recommendation, 20; presentation and documentation, 10.

10.2 Project 20: Expected Loss and Bondholder Recovery

Scenario. A lender is comparing two speculative-grade bonds with different default probabilities, seniority, collateral, and prices.

Objectives

- Estimate simple one-year or cumulative expected credit loss using transparent assumptions.
- Analyze how seniority, collateral, covenants, and capital structure affect recovery.
- Compare expected loss with the observed spread while recognizing liquidity and risk premiums.

Data sources. Issuer capital-structure and covenant disclosures from **SEC EDGAR** or **SEDAR+**; bond prices and trades from **FINRA**; public default and recovery studies from rating agencies; **FRED** high-yield spread indexes; instructor-provided default tables if reports are paywalled.

Deliverables. Capital-structure diagram; default-probability and recovery assumptions; expected-loss scenarios; spread-versus-loss comparison; a creditor preference with limitations; and a 10–12 minute team presentation.

Rubric (100 points). Capital-structure research, 20; expected-loss calculations, 25; recovery reasoning, 25; spread comparison and judgment, 20; presentation and citations, 10.

11 Lecture 11: Interest-Rate Options

11.1 Project 21: Cap, Floor, or Swap for a Borrower?

Scenario. A floating-rate borrower wants protection against rising rates but is uncertain whether to buy a cap, enter a fixed-for-floating swap, or remain unhedged.

Objectives

- Compare payoff asymmetry, premium, opportunity cost, and accounting or liquidity considerations.
- Evaluate the alternatives under at least four future-rate scenarios.
- Recommend a hedge consistent with a stated risk tolerance and budget.

Data sources. **CME Group** SOFR futures and options information; current or instructor-provided cap and swap quote snapshots; SOFR from the **Federal Reserve Bank of New York** or **FRED**; a public company's floating-rate debt disclosure for realistic exposure terms.

Deliverables. Exposure profile; cap, swap, and unhedged payoff table; rate-scenario cost comparison; qualitative volatility discussion; a hedge recommendation and trigger; and a 10–12 minute team presentation.

Rubric (100 points). Exposure and instrument setup, 20; payoff and scenario accuracy, 30; option-versus-swap reasoning, 20; recommendation and risk tradeoffs, 20; presentation and source note, 10.

11.2 Project 22: Option on Treasury Futures Event Hedge

Scenario. A bond manager expects substantial rate volatility around a policy or inflation release but wants to limit downside without giving up all upside.

Objectives

- Choose a call, put, or simple option spread on a Treasury futures contract.
- Map the option payoff to the manager's underlying duration exposure.
- Compare the option hedge with a linear futures hedge across price and yield scenarios.

Data sources. **CME Group** Treasury futures and options contract specifications, settlements, and delayed quotes; economic-release calendars from the **Federal Reserve**, **BLS**, or **Bank of Canada**; instructor-provided option-chain snapshots where historical chains are unavailable.

Deliverables. Exposure and event definition; contract and strike selection; expiration payoff diagram; premium-inclusive scenario table; comparison with a futures hedge; post-event evaluation plan; and a 10–12 minute team presentation.

Rubric (100 points). Event and exposure framing, 15; contract and strike selection, 25; payoff and premium calculations, 25; futures comparison and hedge judgment, 25; presentation and reproducibility, 10.